Jialing Wu

The Ohio State University,
Department of Engineering Education

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Education

The Ohio State University, Department of Engineering Education — Expected May 2028
Ph.D. in Engineering Education
Minor in Computer Science and Engineering — AI Track
Advisor: Dr. Adam Carberry

Vanderbilt University, Peabody College — May 2024
Master of Education, International Education Policy and Management

Shanghai University — July 2017
Bachelor of Engineering, Mechanical Engineering

Research Interests

Methodological Core: AI and computational methods (machine learning, LLMs, text analysis, network analysis, GIS, person-centered methods)
Application Areas: Educational research, teaching practice, and student learning; engineering education; equity and access

Publications

Peer-Reviewed Journal Articles

[22] **Wu, J.**, Shehab, F., & McGuire, L. (in press). Too deep or too shallow: Reflections on methods training from the perspective of engineering education students. *Teaching Educational Research Methods*. https://doi.org/10.5149/term.5040

[21] Yan, J., Wei, X., & **Wu, J.** (2026). Curriculum- and grade-level AI policy implementation: A comparative analysis of North Carolina and California. *Educational Policy*. https://doi.org/10.1177/08959048261481543

[20] **Wu, J.**, Dalal, M., Figard, R., & Carberry, A. R. (2026). Understanding transgender and gender-nonconforming students' experiences in precollege engineering education: A latent profile analysis [Invited contribution based on ASEE PCEE Division Best Paper award]. *Journal of Pre-College Engineering Education Research (J-PEER)*, *16*(1), Article 3. https://doi.org/10.7771/2157-9288.1515

Editorial

[19] Darling-Aduana, J. S., Heinrich, C., Noonan, J., **Wu, J.**, & Enriquez, K. (2026). Failing to learn from failure in online credit recovery assessments [Policy Insights]. *Journal of Policy Analysis and Management*, *45*(3), e70120. https://doi.org/10.1002/pam.70120

Book Chapters

[18] Yan, J., **Wu, J.**, & Wei, X. (2026). Silent policies: How GenAI guidelines in U.S. higher education overlook international students' experiences. In A. Chen (Ed.), *Generative AI for supporting international student well-being in higher education* (pp. 95–132). IGI Global. https://doi.org/10.4018/979-8-3373-6571-8.ch004

Conference Papers

[17] **Wu, J.**, Ren, S., & Wang, S. (2026). Using bipartite network analysis to examine global collaboration between engineering and STEM education (2014–2024). *2026 ASEE Annual Conference & Exposition Proceedings*, Charlotte, NC. https://doi.org/10.18260/1-2--61184

[16] **Wu, J.**, & Wang, S. (2026). What does the "person-centered quantitative approach" mean in practice? A literature review in STEM education research [Work-in-progress paper]. *2026 ASEE Annual Conference & Exposition Proceedings*, Charlotte, NC. https://doi.org/10.18260/1-2--60763

[15] **Wu, J.**, & Carberry, A. R. (2026). Integrating robotics and engineering education: Initial validation testing of an SCCT- and CT-based survey instrument [Work-in-progress paper]. *2026 ASEE Annual Conference & Exposition Proceedings*, Charlotte, NC. https://doi.org/10.18260/1-2--60118

[14] **Wu, J.**, & Melo de Lyra, M. V. (2026). Exploring perspectives on pre-college robotics education: A social media–based topic and emotion analysis [Work-in-progress paper]. *2026 ASEE Annual Conference & Exposition Proceedings*, Charlotte, NC. https://doi.org/10.18260/1-2--59786

[13] Melo de Lyra, M. V., **Wu, J.**, & Carberry, A. R. (2025). Exploring early-career engineering faculty experiences teaching using social cognitive theory. *2025 IEEE Frontiers in Education Conference (FIE)*, Nashville, TN, pp. 1–10. https://doi.org/10.1109/FIE63693.2025.11328485

[12] Dalal, M., **Wu, J.**, & Iqbal, A. (2025). Overcoming challenges in robotics education: Examining teacher facilitators and barriers. *Research in Engineering Education Symposium (REES)*, Johor Bahru, Malaysia.

[11] **Wu, J.**, Dalal, M., & Carberry, A. (2025). Advancing equity: Exploring the experiences of transgender and gender non-conforming students in a pre-college engineering course [Work-in-progress paper]. *2025 ASEE Annual Conference & Exposition Proceedings*, Montreal, QC, Canada. https://doi.org/10.18260/1-2--57577

[10] Wei, X., **Wu, J.**, & Klein-Gardner, S. S. (2025). Parents' knowledge, attitude, and behavior on pre-college engineering education course [Work-in-progress paper]. *2025 ASEE Annual Conference & Exposition Proceedings*, Montreal, QC, Canada. https://doi.org/10.18260/1-2--55568

[9] Lachapelle, C. P., Dalal, M., McKeown, K., & **Wu, J.** (2025). Curriculum design for all learners. *2025 ASEE Annual Conference & Exposition Proceedings*, Montreal, QC, Canada. https://doi.org/10.18260/1-2--56190

[8] **Wu, J.**, & Dalal, M. (2024). High school students' perspectives on pre-college engineering education courses (Fundamental). *2024 ASEE Annual Conference & Exposition Proceedings*, Portland, OR. https://doi.org/10.18260/1-2--47527

[7] **Wu, J.**, Léger, N., & Klein-Gardner, S. (2024). High school students' perspectives on mathematical modeling in the engineering design process (RTP). *2024 ASEE Annual Conference & Exposition Proceedings*, Portland, OR. https://doi.org/10.18260/1-2--47528

Conference Presentations

[6] **Wu, J.**, & Li, D. (2026, November 5–7). Teachers as AI policy actors: Focus group evidence from a K–12 robotics consortium [Poster presentation]. *Association for Public Policy Analysis & Management (APPAM) Annual Fall Research Conference*, Boston, MA.

[5] Wang, J., **Wu, J.**, & Cheng, Y. (2026, July 28–31). Cross-cultural evaluation of LLMs: A mixed-method study of Chinese-to-English movie title translation [Poster presentation]. *12th International Conference for Computational Social Science (IC2S2 2026)*, Burlington, VT.

[4] Wei, X., & **Wu, J.** (2026, April 8–12). Navigating AI in higher education: Student views on professors' attitudes toward AI tools in the classroom analyzed through sentiment analysis and topic modeling [Conference presentation]. *American Educational Research Association (AERA) Annual Meeting*, Los Angeles, CA.

[3] **Wu, J.**, & Yan, J. (2026, March 28 – April 1). Comparing high school accessibility in California and New York: A GIS-based assessment [Conference presentation]. *Comparative and International Education Society (CIES) Annual Conference*, San Francisco, CA.

[2] Wei, X., **Wu, J.**, Klein-Gardner, S., & Dalal, M. (2025, November 10–14). Evaluating fidelity of teaching in pre-college engineering: A case study of curriculum implementation [Poster presentation]. *Evaluation 2025*, Kansas City, MO. eval25.eventscribe.net

[1] Darling-Aduana, J. S., Heinrich, C., Noonan, J., **Wu, J.**, & Enriquez, K. (2025, April 23–27). Failing to learn from failure: The facade of online credit recovery assessments [Conference presentation]. *American Educational Research Association (AERA) Annual Meeting*, Denver, CO. https://doi.org/10.3102/ip.25.2182630

Manuscripts Under Review / In Preparation

[2] **Wu, J.**, Wei, X., & Yan, J. (under review). Engineering education deserts in the United States: A geographic random forest analysis of broad-access engineering programs in higher education. *AERA Open*.

[1] Berhane, B., Dalal, M., Ross, L., **Wu, J.**, & Wang, J. (under review). Examining student- and school-level outcomes in pre- and post-survey data from a pre-college engineering program. *Journal of Engineering Education*.

Grants, Fellowships, and Awards

Grants

Sponsored Graduate Student Registration (awarded), AERA Division D (Measurement & Research Methodologies) — 2026

Smith Travel Grant ($1,000; awarded), Department of Engineering Education, OSU — 2025, 2026

Career Development Grant ($350; awarded), Council of Graduate Students, OSU — 2025

Fellowships and Scholarships

Frédéric Bastiat Fellowship ($5,000), Mercatus Center at George Mason University — 2025–2026

Scholastic Achievement Scholarship ($26,500), Vanderbilt University, Peabody College — 2022–2024

University Scholarship for Academic Excellence (¥1,000), Shanghai University — 2014, 2015

Honors and Awards

Best Paper Award, 2nd Place, Pre-College Engineering Education Division, ASEE — transgender and gender-nonconforming students in pre-college engineering — 2025

Professional Experience

Research Experience

The Ohio State University, Columbus, OH — August 2024 – Present
Graduate Research Associate, Department of Engineering Education
(with Dr. Adam Carberry)

The e4usa+FIRST program pairs the e4usa curriculum with FIRST Robotics for high school students.

- Lead the development of survey instruments grounded in Social Cognitive Career Theory (SCCT) and computational thinking to assess student self-efficacy, identity, and skill development.
- Collect pre-test data via Google Forms for initial validity checks of the survey instrument.
- Prepare and submit IRB applications and protocols across multiple high school program sites.
- Conducted qualitative analysis of teacher interviews on robotics curriculum implementation, identifying facilitators and barriers; co-authored methods and results sections.

Vanderbilt University, Nashville, TN June 2023 – July 2024
Graduate Research Assistant, School of Engineering
(with Dr. Stacy Klein-Gardner, Dr. Medha Dalal)
e4usa is a national initiative introducing engineering design principles to high school students.

- Investigated high school students' perspectives on engineering courses through qualitative analysis of post-test responses; developed codebooks and conducted pre/post statistical comparisons across cohorts.
- Analyzed focus group data on mathematical modeling tools through cognitive load theory.
- Co-authored two ASEE 2024 conference papers; led research group meetings and managed timelines.
- Revised Unit 2 curriculum (water filters → wind turbines), redesigning lessons and MATLAB code.
- Drafted facilitator memos for summer teacher PD programs and managed Canvas course materials.

Teaching Experience

The Ohio State University, Columbus, OH August 2026 – May 2027
Graduate Teaching Associate, Department of Engineering Education
ENGR 1181: Fundamentals of Engineering I

- Facilitate four weekly laboratory sections for first-year engineering students, delivering short instructional presentations and guiding 80-minute hands-on engineering activities.
- Coordinate with faculty instructors and undergraduate teaching assistants to implement labs, manage attendance and team processes, and apply course policies consistently across sections.
- Assess laboratory assignments and support students through office hours, make-up arrangements, individualized feedback, and accessible instructional practices.

Industry Experience

Howard Hughes Medical Institute (HHMI), Chevy Chase, MD May 2024 – August 2024
Assessment Intern, Science Education (BioInteractive Team)

- Developed an HTML codebook (9 chapters, 25 sections) guiding accessible Canvas course design.
- Conducted UDOIT accessibility audits and authored a guideline handbook for content editors.

Shanton Way Executive Search, Shanghai, China July 2017 – July 2022
Consultant

- Executive recruitment for clients in international K–12 education, consulting, finance, and technology sectors.
- Sourced and managed a candidate pool of 1,500+ bilingual professionals across international schools.

Workshops and Facilitation

From Reddit to Policy Insights: Using AI to Summarize Public Opinion on Robotics Courses November 2026
Facilitator, AI Peer Learning Lab, "From Text to Evidence: AI-Assisted Analysis and Measurement," 2026 APPAM Fall Research Conference, Boston, MA

Multilevel Modeling October 2026
Organizer and Presenter, OTTER (Open Toolkit for Training in Education Research), The Ohio State University

Training and Certificates

College and University Teaching Graduate Certificate In Progress
The Ohio State University — Selected coursework: College Teaching; Learning Theory, Pedagogy, and Assessment in Engineering Education

Summer Institute in Computational Social Science 2025
University of Rochester — Topics: LLMs, network analysis

ICPSR Summer Program 2024
University of Michigan — Topics: Python programming, R programming, text analysis

Service and Leadership

Professional Service

Reviewer; Journal of Engineering Education 2026

Reviewer; AERA Annual Meeting 2026

Reviewer; ASEE Annual Conference & Exposition 2025, 2026

Discussant; APPAM Annual Fall Research Conference 2026

Student Session Moderator; ASEE Annual Conference & Exposition 2026

Reviewer; SACNAS NDiSTEM Conference 2025

Community Service

Institutional Coordinator; Engineering Education Graduate Student Exchange Program, OSU 2025–Present

Graduate Ambassador; Department of Engineering Education, OSU 2025–Present

Graduate Student Host; Faculty Candidate Visits, Department of Engineering Education, OSU 2024–2026

Volunteer; Florence Nightingale Day for Statistics and Data Science, OSU 2025

Founding Member and Treasurer; iMAP, Vanderbilt University 2023–2024

Mentor; Peabody Peer Career Mentor Program, Vanderbilt University 2023–2024

Mentor; Peabody International Students and Affairs, Vanderbilt University 2023–2024

Migrant Children Program Coordinator and Teacher; AIQUZHI Volunteer Group 2017–2022

Professional References

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